TRACE-RPG: A Trace-Linked Symbolic Commit Gate for Generated Events in an Interactive Game World

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***Abstract*—Large language models can generate expressive quests and non-player-character dialogue, but fluent or schema-valid text does not establish that a proposed event is executable, authorized, or consistent with canonical game state. We present TRACE-RPG, a symbolic commit gate that isolates untrusted proposals from state mutation. A type-strict parser, externally supplied action policy, and six deterministic state-relative predicates decide every admitted transition; rejected candidates may enter bounded repair, and accepted candidates are revalidated before mutation. The counterexample-guided operator ρ reads only the prior candidate and structured validation errors, never authoritative state. Unkeyed SHA-256 content links, state-semantic replay, episode-continuity checks, and assignment-complete outcome accounting make the resulting traces auditable within that integrity boundary. We keep three evidence lanes separate: a deterministic offline conformance pilot over one authored world, a single-model live screening extension, and an engineering-only closed-world KG/ontology link-proposal simulation that has no runtime authority. The offline lane establishes encoded-field mechanism conformance only. At $K=1$, the live lane observed 5/5 guided commits versus 0/5 blind commits only in a deliberately repairable policy-blind regime; the other regimes were null, so this result remains pilot-only. No lane establishes population efficacy, model superiority, player experience, affect, runtime retrieval or memory benefit, or commercial-engine performance.**

***Index Terms*—game artificial intelligence, interactive narrative, symbolic validation, constrained generation, runtime enforcement, reproducibility**

I. Introduction

INTERACTIVE generation is more than a language task. A candidate event may be natural yet require an absent item, assert a fact unknown to its speaker, reveal forbidden quest information, introduce an unauthorized effect, or regress a quest stage. Executable text environments make the distinction between a plausible utterance and a valid transition explicit [1], [2]. Grounded dialogue and personalized quest systems further show the value of quest, entity, and graph context [3], [4], but context does not itself authorize canonical mutation.

Structural generation constraints solve an adjacent problem. Grammar-constrained decoding, language-model query languages, and incremental parsers can restrict generated structures [5], [6], [7]. A syntactically admissible event can nevertheless be false relative to the current world. Retrieval and memory can supply useful evidence [8], [9], [10], but retrieved content is also not a state-transition authority.

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This paper presents TRACE-RPG, an implementation that combines these concerns as an auditable transaction protocol. Its current contributions are:

1) **C1—Contracts:** *versioned trust-boundary contracts* for state, policy, candidate, record, and game bridge, with a type-strict parser that rejects ambiguous known-field types and unknown candidate keys at proposal and replay boundaries;
2) **C2—Controller:** *a validate–repair–commit controller* with six state-relative predicates, bounded repair, unchanged fallback, and defensive pre-commit revalidation;
3) **C3—Audit:** *an audit-evidence layer* using unkeyed SHA-256 content links, state-semantic replay, and episode-continuity checks;
4) **C4—Harness:** *an offline conformance harness* with frozen assignment keys, classified terminal failures, and assignment-complete accounting, including failures for which no proposal trace can exist; and
5) **C5—Repair:** *a counterexample-guided operator* $\rho(a, E)$ that reads only the prior candidate and structured errors, compared with blind unchanged retry and a state-reading oracle upper bound across frozen repairability classes.

A separate structured crosscheck binds C1–C5 to prior-work topics, citation keys, claim IDs, repository evidence, evidence lanes, and inference ceilings. Deterministic parser, state-isolation, repair, fault-injection, and manifest fixtures provide the exact-count evidence for these five implementation contributions. The evidence map has three non-interchangeable lanes. First, the empirical center is a deterministic conformance pilot over one authored world state; it measures encoded parser, validation, repair, replay, and accounting behavior, not population efficacy. Second, a five-call-per-cell live screening extension probes repair transfer with one hosted proposer; its unpinned revision, uncontrolled seed labels, and deliberately constructed states cap it at pilot-only. Third, a closed-world KG/ontology simulation evaluates authored typed-link proposals but neither implements runtime retrieval nor supports a `C-RESULT-*` claim. A retained Godot 4.7.1 policy-mirror trace and render replay add engine-local conformance evidence only. In four fixture runs, the largest of five sampled headless frame deltas was always the first (98.760–116.667 ms), missing the 16.7 ms budget; these startup-sensitive traces are not a stable frame benchmark.

There are no participants, affect or runtime-retrieval experiments, live Python–Godot authorization round trips, or representative end-to-end gameplay performance measurements.

II. Related Work and Research Gap

A. Interactive Narrative and Game Agents

KNUDGE evaluates branching NPC dialogue grounded in quest and entity specifications [3]. Knowledge-graph-assisted quest generation [4], memory–reflection–planning agents [11], and trainable role-playing agents [12] address complementary aspects of grounded and believable behavior. Player-facing studies evaluate preference and perceived dialogue quality [13], [14]. Yin *et al.*'s 2026 online record has an issue assignment that was future-dated at our metadata audit and must be rechecked at submission [15]. A 2025 preprint reports role-sensitive stability–improvisation trade-offs [16]; we use it only as recent motivation. These works motivate separating hard admissibility from optional naturalness and player-experience constructs.

Experience-driven content generation [17] and observer-labelled engagement [18] motivate a future adaptation track. A recent preprint reports limitations in vision-language-model engagement inference [19]. Accordingly, affect remains non-authoritative and is absent from the present implementation evidence.

B. Environments and Benchmarks

TextWorld and ScienceWorld expose explicit interactive state [1], [2]; MineDojo adds open-ended, knowledge-rich embodied tasks [20]. General Video Game AI separates agent, game, and content tracks [21]. AgentBench studies agents across heterogeneous environments [22], AgentBoard adds process-level diagnostics [23], SOTOPIA evaluates social interaction [24], and BALROG evaluates long-horizon language and visual game reasoning [25]. Automated play-style agents provide a complementary testing precedent, not a substitute for human evaluation [26]. These benchmarks inform our future evaluation design but do not prescribe a transactional authorization boundary.

C. Constraints, Neuro-Symbolic Validation, and Retrieval

Constrained decoders improve output structure [5], [6], [7]; TRACE-RPG instead treats structure as an input condition to independent state-relative authorization. Two direct game-specific comparators span complementary boundaries. PANGeA combines memory, validation, a REST service, and Unity integration, but its documented validation asks an LLM to judge and iteratively refine content against designer rules [27]. IVIE, currently a preprint associated with non-archival ICCC 2026 material, incrementally generates interactive-fiction worlds with symbolic validation [28]. A 2026 system in the same lineage restricts LLMs to selecting authored world-state transformations and reports an exploratory eight-participant study of player expression [29]; it maintains consistency by construction but provides no transactional gate, audit-linked records, or repair-operator evidence. Neither comparison implies superiority: each reinforces that validity is only validity for the objectives and predicates its system encodes.

G-Retriever, HippoRAG, and Mem0 address relevant-subgraph retrieval and long-term memory [8], [9], [10]; Generative Agents and Character-LLM address behavioral memory and persona [11], [12]. TRACE-RPG may later consume such context, but denies it direct commit authority. Retrieval and temporal-memory benefits are not measured here.

D. Auditability and Reproducibility

Process metrics motivate retaining intermediate failures [23]; robust evaluation cautions against overinterpreting sparse stochastic runs [30]. Reproducibility and environmental reporting motivate frozen artifacts and explicit measurement boundaries [31], [32]. Human and LLM evaluation require construct, rater, and bias controls [33], [34], [35], [36], while future non-inferiority and participant analyses require justified margins and random-effects structures [37], [38].

The bounded novelty is the implemented combination, not the invention of its individual parts. Action interposition is long established. Narrative mediation intercepts a user or agent action before it executes and either blocks it or accommodates it by revising the plan [39]; classical planning defines applicability through preconditions and effects [40]; runtime enforcement wraps an untrusted actor in a monitor that admits only permitted transitions [41]; authored-logic interactive drama governs which utterances a character may make, in this journal's predecessor title [42]; and belief-aware narrative planning reasons over what characters know, which is the concern our knowledge and disclosure predicates encode far more coarsely [43]. TRACE-RPG does not claim to have invented interposition. C1–C2 place strict contracts and a validate–repair–commit controller over this lineage; C3–C4 combine checksum-linked terminal records, state-semantic replay, episode continuity, and assignment-complete accounting that retains cases for which no proposal trace can exist; C5 isolates a repair operator that consumes only typed pre-execution validator errors. We do not infer that a feature absent from a report is absent from every implementation.

III. Problem Definition and Guarantee Boundary

For exposition, the controller input used by validation at step t is decomposed as

$$c_t = (G_t, q_t), \tag{1}$$

where G_t denotes typed snapshot fields and q_t denotes the externally authored action-policy, disclosure, and quest-stage constraints stored with that snapshot. Terminal record history $h_{<t}$ is maintained separately from the `WorldState` snapshot; no validation predicate reads it, and the implementation does not reconstruct state from the log. Retrieval, narrative scores, model rationales, memory summaries, and any future affect estimate are non-authoritative proposal context.

An untrusted proposer emits candidate a_t. Strict parsing first enforces the shared candidate-key contract, including rejection of unknown top-level fields, and establishes the declared schema. The validator returns

$$V(c_t, a_t) = (v_{\text{policy}}, v_{\text{pre}}, v_{\text{reach}}, v_{\text{know}}, v_{\text{disc}}, v_{\text{quest}}, E_t), \tag{2}$$

where E_t is a structured error set. Mutation is

$$c_{t+1} = \begin{cases} T(c_t, a_t), & \bigwedge_i v_i = 1, \\ c_t, & \text{otherwise.} \end{cases} \tag{3}$$

The controller asserts the following implementation invariants under its encoded contracts; the pilot tests their mechanisms with deterministic valid, invalid, repair, fallback, replay, and fault-injection fixtures rather than presenting general theorems. **I1 (exhausted-failure immutability):** if no returned candidate through budget K validates, or proposal/controller execution fails before commit, the returned state equals the supplied prior state. **I2 (bounded recorded attempts):** provided every proposal and repair callback returns, repair budget $K \geq 0$ admits at most $K + 1$ recorded candidate attempts; a successful commit adds one defensive validation immediately before mutation. The runtime does not enforce callback deadlines, so I2 is not a wall-clock termination guarantee. **I3 (effect authorization):** an accepted candidate contains no declared fact effect in `CandidateAction.effects` outside `ActionPolicy.allowed_effects`, and any quest-stage target is explicitly authorized by that policy. **I4 (state-semantic replay):** with the frozen schemas and deterministic transitions, an accepted trace reconstructs its stated terminal state.

These asserted invariants assume correct policy authorship, complete and faithful extraction into declared candidate fields, deterministic software versions, returning callbacks, and single-process file ownership. State-semantic replay revalidates the recorded structured candidate and transition; it does *not* reproduce or verify the provenance, reasoning, prompts, or stochastic generation of intermediate repairs. Natural-language implications omitted from structured fields remain outside the guarantee.

IV. TRACE-RPG Architecture and Algorithms

A. Trust Boundary and Contracts

Fig. 1 places the model or recorded adapter outside the canonical owner. The parser rejects ambiguous numeric and collection types for known fields and rejects unknown top-level candidate fields under the shared key contract. The action policy is supplied separately from the candidate and defines required preconditions and allowed effects. The game bridge is a versioned event interface that may carry observations, proposals, validations, rejections, fallbacks, or commits; only a valid commit authorizes canonical mutation. This decouples interfaces but does not yet demonstrate portability to a commercial engine. Table I summarizes the encoded rejection boundary; it is not a claim of semantic completeness.

B. Validate–Repair–Commit

Fig. 2 gives the controller's dependency-free pseudocode. "Record" denotes an append to the in-memory trace representation; persistent writing occurs only after terminal consistency checks.

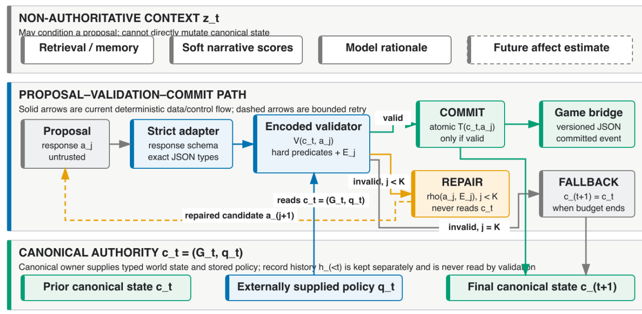


Fig. 1. Trust boundary. Retrieved context and generated candidates remain non-authoritative until known-field type checks and state-relative validation complete.

TABLE I
Encoded Validator Boundary

Predicate	Rejects	On failure
Action policy	unknown action/type	unchanged
Precondition	absent/false requirement	unchanged
Reachability	inaccessible object	unchanged
NPC knowledge	undeclared known fact	unchanged
Disclosure	forbidden fact	unchanged
Quest	ineligible/regressive stage	unchanged

Validation errors are exposed to a repair callback as structured error codes. Repair never directly mutates state, and exhaustion returns an unchanged deterministic fallback. Fig. 3 shows terminal paths. The pilot compares two implemented callbacks. The *counterexample-guided operator* consumes only the prior candidate a and the validator's structured error set E, never the authoritative state:

$$\rho(a, E) = a \oplus \bigcup_{e \in E} \delta(e), \tag{4}$$

where each $\delta(e)$ is the deterministic per-error-class edit of Table II and $\oplus$ applies the resulting field edits. Each edit touches at most one candidate field per error entity, so the number of changed fields per attempt is bounded by $|E|$; applying ρ twice with the same error set is a fixed point; and an entity simultaneously demanded and rejected on the same field is declared irreparable and left untouched. The *state-reading reference callback* instead discards the error set and reconstructs the policy-governed fields (preconditions, effects, and both quest-stage fields) directly from the authoritative state; it deliberately never edits declaration-integrity fields (required objects, used facts, disclosed facts), because undeclaring a dependency or disclosure would launder the violation past the gate rather than repair the candidate. It is an oracle upper bound over state-readable repairs, and its outcomes must not be read as evidence about any deployable repair method.

C. Integrity, Semantic Replay, and Accounting

Each result row carries an unkeyed SHA-256 *integrity checksum*; proposal outcomes additionally link to a trace checksum. These hashes detect accidental or tested content

$a \leftarrow$ PROPOSE(c); $j \leftarrow 0$
repeat
  $(ok, E) \leftarrow V(c, a)$; RECORD$(a, E)$
  if ok **then**
    assert $V(c, a).ok$; **return** COMMIT(c, a)
  if $j = K$ **then return** FALLBACK(c)
  $a \leftarrow$ REPAIR(c, a, E); $j \leftarrow j + 1$
until terminal

Fig. 2. Validate–repair–commit procedure. At most $K+1$ candidate attempts are recorded; the successful path performs a defensive validation before mutation.

TABLE II
PER-ERROR-CLASS EDITS OF THE GUIDED OPERATOR ρ

Validator error class	Edit $\delta(e)$ on the candidate
Policy precondition omission	insert the named predicate
Policy effect omission	insert the named effect
Policy effect violation	drop the named effect
Unsatisfied precondition	drop the named predicate
Unauthorized stage mutation	clear `quest_stage_effect`
Unknown action type	no-op (no registered alternative)
Unreachable object; knowledge, disclosure, and stage classes	no-op (candidate-local edit would launder or needs state)

changes but are not signatures, message-authentication codes, writer identity, or protection against an actor able to rewrite content and checksums. Semantic replay goes beyond byte integrity by strictly parsing the prior state and every recorded candidate/validation pair, requiring every nonfinal attempt to be invalid, recomputing the terminal transition, and comparing the stated final state. Episode replay also requires state continuity between adjacent JSONL records. Replay does not authenticate or re-execute the repair-generation step that produced candidate $j+1$.

The terminal record classes are commit, symbolic fallback, adapter failure, and controller failure. A controller-failure row remains in the frozen assignment denominator and leaves state unchanged, but it may legitimately lack a completed proposal trace because no candidate outcome was produced. Accordingly, result completeness does not imply trace existence for every terminal row. The pre-execution assignment key contains run, arm, scenario, seed, model, revision, controller-configuration hash, assignment-input hash, and prior-state hash. Aggregation rejects duplicate, missing, or unexpected keys and separately reports symbolic, adapter, and controller failures.

V. EVIDENCE LANES AND BOUNDARIES

A. *E1 Offline Conformance Design*

The pilot asks whether: (PQ1) valid candidates commit while named invalid-policy cases remain unchanged; (PQ2) rejection-only, blind unchanged retry, the counterexample-guided operator ρ, and the state-reading reference callback produce the expected outcomes on authored cases in every frozen repairability class; (PQ3) frozen checksum, state-semantic, linkage, and continuity mutations are rejected by their designated check operations, while an injected second-file append failure triggers the expected pair-write rollback check; (PQ4) strict adapters and assignment manifests retain exactly one classified terminal row per assigned case or fail closed;

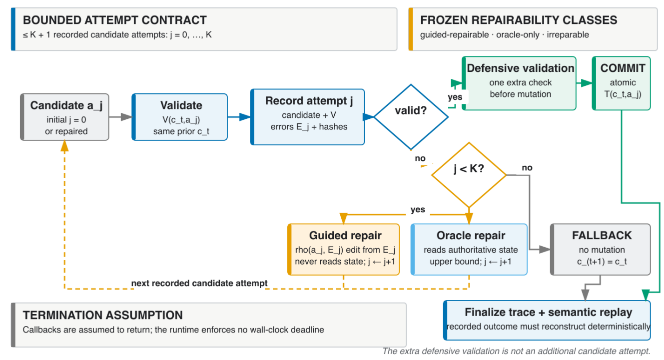


Fig. 3. Proposal, validation, bounded repair, fallback, commit, and replay states. Controller failure is terminal and may have no completed proposal trace.

and (PQ5) proposal and replay parsers both reject a complete otherwise-valid candidate carrying one unknown top-level key.

Cases are purposively designed conformance fixtures, not samples from a model or game population. The four domains are validator/state isolation, repair-arm comparison, record/trace fault injection, and adapter/accounting contracts. Every repair fixture declares a frozen repairability class before execution: *guided-repairable* (the structured error payload alone carries sufficient information for ρ), *oracle-only* (repair additionally requires reading the authoritative state), or *irreparable* (no implemented arm may commit), and each class's expected outcome per arm is part of the designed-fixture contract. Unique cases, rather than repeated deterministic seeds, form descriptive denominators. All expectations are specified before execution.

B. *E1 Measures and Frozen Artifacts*

Measures are exact counts: expected/observed validator-class agreement; rejected or controller-failure cases with state mutation; commit and attempt outcomes by repair arm and repairability; rejected/prespecified detectable faults by designated operation; replayed/committed traces; rejected/injected manifest violations; and proposal/replay rejection agreement on the prespecified unknown-key negative regression. The last measure is deterministic parser-contract parity, not an empirical safety outcome. Exception class alone does not identify a stable detector layer, so no detector-attribution result is reported. No p-values or population confidence intervals are attached to these designed fixtures. The adapter cases carry synthetic latency and token constants that the input schema pins with JSON-Schema `const`; they exercise accounting-field propagation only and are not measurements of any provider, runtime, or device.

The release packet freezes fixture and assignment manifests, schemas, a result JSON artifact with embedded outcome and trace records, environment and code revision, generated tables, and checksums. Reporting follows reproducibility and measurement-boundary guidance [31], [32].

C. **E2** Live RQ2 Screening

The live extension compares blind unchanged retry with ρ at matched K=1. For each cell, five seed-indexed Codex CLI calls request one candidate from `gpt-5.6-sol`; the same candidate is then supplied to both arms, eliminating within-call proposal differences. Conditions vary policy visibility and whether a trivial zero-effect route exists. The promoted packet retains the corrected current cells and the superseded v1 diagnostic, binds every JSON/JSONL receipt by SHA-256, and labels the evidence `screening-pilot-only`. Seed labels do not control the hosted sampler, the hosted revision is unpinned, token accounting is unavailable, and the constructed states are not population samples. We therefore report exact descriptive counts and failure classes without p-values, confidence intervals, or model ranking.

D. **E3** KG/Ontology Link-Proposal Simulation

`SL-KG-ONTOLOGY-SIM-001` is an engineering-only closed-world lane, separate from the Graphify document-navigation graph and the authoritative typed `WorldState`. It exports 43 repository-local OKF nodes and 106 reference edges, overlays 24 reviewed typed edges, checks type/domain/range integrity and six source-scoped competency questions, and materializes a non-authoritative SQLite mirror. Seven fixed strategies score six authored queries with five candidates each. The retained `S2-typed-lexical-loose` strategy recovered 6/6 authored holdouts ($P = R = F_1 = 1.000$, realistic-tie $\mathrm{MRR} = 0.944$, Brier $= 0.131$, $\mathrm{Sem@3} = 1.000$), whereas the degree baseline recovered none. Machine-generated JSON, TSV, TeX, and SVG artifacts carry the full formulas and trials. These are construction results, not OWL/SHACL conformance, semantic completeness, runtime retrieval, long-horizon contradiction evidence, or support for any `C-RESULT-*` claim.

VI. Pilot Results

A. Declared-Field Gate Conformance

The pilot loader admits a fixture set only when it isolates every implemented validator code exactly once alongside one valid control, so the 13/13 agreement and the observation of all 12 implemented error codes are *construction invariants of the harness*, not measured success rates. What the run adds is the part the loader does not enforce: each of the 12 isolated negative fixtures showed observed agreement with its authored expected code rather than reaching a different one, and every noncommit path preserved the complete canonical state. This is implementation conformance to an authored structured-field oracle over a single world state, not accuracy against independent semantic labels.

B. Repair-Arm Comparison on Designed Fixtures

The 12 initially invalid designed cases are partitioned into frozen repairability classes: 5 guided-repairable, 1 oracle-only, and 6 irreparable. Rejection-only and blind unchanged-candidate retry each committed 0/12. The counterexample-guided operator ρ committed 5/5 guided-repairable cases and, by design, 0/1 oracle-only and 0/6 irreparable cases; every ρ commit edited a mean of 1.0 candidate fields, and every failed arm execution left the canonical state unchanged. The state-reading reference callback committed 5/5 guided-repairable and 1/1 oracle-only cases, with 0/6 irreparable. These exact counts separate the three mechanisms on the designed fixtures: blind retry never converts a counterexample into a commit, ρ converts exactly the cases whose error payload carries sufficient information, and the oracle additionally converts the case requiring state knowledge. This is operator conformance on frozen fixtures, not repair quality of any live model.

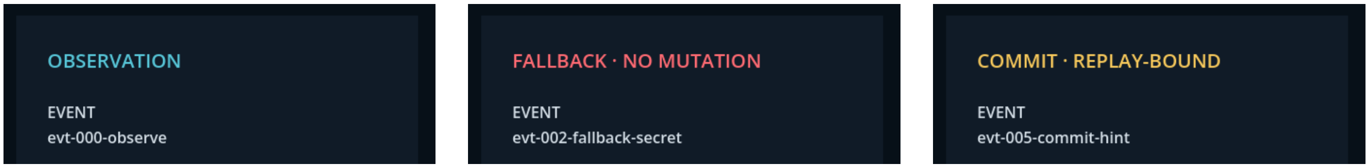
TABLE III
Repair Arm × Repairability Class, Exact Counts (Designed Fixtures)

Arm	Info source	G-rep.^{f}	O-only^f	Irrep.^{f}	Edits^e
Rejection-only (K=0)	none	0/5	0/1	0/6	—
Blind unchanged retry	none	0/5	0/1	0/6	—
Counterexample-guided ρ	error set E	5/5	0/1	0/6	1.0
State-reading oracle	state+policy	5/5	1/1	0/6	1.0

C. Integrity, Boundaries, and Assignment Accounting

All 10 faults prespecified as detectable were rejected by their designated check operations (10/10). Because this pilot does not compare stable typed detector codes, it does not establish detector-layer attribution. Separately, the known provenance-boundary fixture substituted a different invalid precursor before the same recorded valid repair, recomputed the checksum, and passed replay as expected (1/1 undetected). Thus, the mechanism does not protect against a writer who can recompute unkeyed checksums, and replay does not authenticate the repair-generation operation. Two open boundary sentinels—unextracted narrative disclosure and simultaneous omission of a required object from candidate and policy—were intentionally accepted at the encoded layer (2/2), with 0 labelled as safety passes. The former unknown-top-level-field acceptance boundary was reclassified after Stage 8 as a closed negative regression: both proposal and replay parsers specifically rejected a complete 12-field candidate carrying one unknown key (1/1). This establishes parser rejection parity, not semantic safety. Among seven adapter/accounting assignments, outcomes were 1 commit, 1 symbolic fallback, and 5 adapter failures. Synthetic telemetry fields were populated and propagated for 2/7 assignments; these are schema-pinned constants verifying accounting-field propagation, not measurements. All 3/3 injected duplicate or missing-assignment guards failed closed. These are raw counts from synthetic offline fixtures.

Tables III and IV report exact designed-fixture counts: the repair table separates the four arms per frozen repairability class, and neither table supports a population or live-model estimate.



(a) Arrival observation (b) Unchanged-state rejection (c) Authorized commit

Panel	Capture	Event	Validation	State relation
(a)	`sl-rc-001-arrival`	`evt-000-observe`	not applicable	snapshot = before
(b)	`sl-rc-002-rejected-secret`	`evt-002-fallback-secret`	invalid	world = before
(c)	`sl-rc-003-authorized-hint`	`evt-005-commit-hint`	valid	world $\neq$ before

Full-resolution artifact pointer: godot-4.7.1-20260813t115916z-sealed-lighthouse-render-v5/rendered-canonical-v1/; filenames are the capture IDs above plus `.png`.

Fig. 4. Illustrative status/event crops from the authored Godot render replay, with manifest metadata. The trace-bound correspondence is not live-transport, performance, usability, immersion, or efficacy evidence.

TABLE IV
FROZEN-PILOT RAW ACCOUNTING

Check	Numerator	Denominator
Gate fixture agreement[c]	13	13
Known-boundary acceptances[a]	2	2
Closed unknown-key boundary rejected[d]	1	1
Detectable integrity faults rejected	10	10
Known integrity boundary replay-accepted[b]	1	1
Adapter commits	1	7
Symbolic fallbacks	1	7
Adapter failures	5	7
Manifest guards rejected	3	3

[a]Boundary acceptances document semantic-extraction and policy-completeness limits; they are not safety passes. [b]Replay does not authenticate or re-execute repair generation. [c]The loader enforces this agreement, so it is a construction invariant rather than a measurement. [d]This is post-Stage-8 parser rejection parity, not semantic-safety evidence. [e]Mean changed candidate fields over committed repairs (minimality proxy); failed arm executions left the state unchanged. [f]Commits/cases. G-rep. = guided-repairable, O-only = oracle-only, Irrep. = irreparable (frozen repairability classes).

D. Live Screening Pilot (Pilot-Only)

A separate screening pilot called gpt-5.6-sol through the Codex CLI. It used 5 seed-indexed proposal calls per cell and repair budget $K = 1$; within each call, blind unchanged retry and ρ received the same initial candidate. The hosted revision was unpinned, token accounting was unavailable, and the seeds did not control the hosted sampler, so the calls are quasi-replicates rather than independent randomized samples. We report neither inferential statistics nor a model ranking.

In Table V, a guided advantage appeared only in the policy-blind `signal-repair-v2` cell, a state deliberately constructed to elicit guided-repairable errors. All 5/5 initial candidates were invalid; ρ committed 5/5, whereas blind retry committed 0/5. The other four current cells showed no guided advantage: three had already-valid initial candidates, and the frozen-base goal-directed cell produced irreparable quest-stage regressions. All 15/15 noncommit arm outcomes in the current cells preserved the prior-state hash. This supports mechanism transfer only in that error regime, not population efficacy, model promotion, or a general sample-efficiency claim. Accordingly, `C-PILOT-007` and `C-PILOT-008` are pilot-only while `C-RESULT-003` remains `TODO-RESULT`. In the superseded v1 diagnostic, guided repair reduced up to 3 co-occurring errors to one irreparable knowledge error but still could not commit, showing that partial per-error repairs do not compose when any residual error is irreparable.

TABLE V
LIVE SCREENING PILOT, EXACT COUNTS (FIVE PROPOSALS PER CELL, K=1)

Base / condition	Initial valid	ρ commits	Blind commits	Initial errors
Frozen / visible	5/5	5/5	5/5	—
Frozen / blind	5/5	5/5	5/5	—
Frozen / goal-blind	0/5	0/5	0/5	QSR
Signal-v2 / visible	5/5	5/5	5/5	—
Signal-v2 / blind	0/5	5/5	0/5	PEO+PEV+PPO

QSR = quest-stage regression, PEO = policy-effect omission, PEV = policy-effect violation, PPO = policy-precondition omission. All counts are screening-pilot-only, without inferential statistics; the superseded v1 diagnostic is excluded.

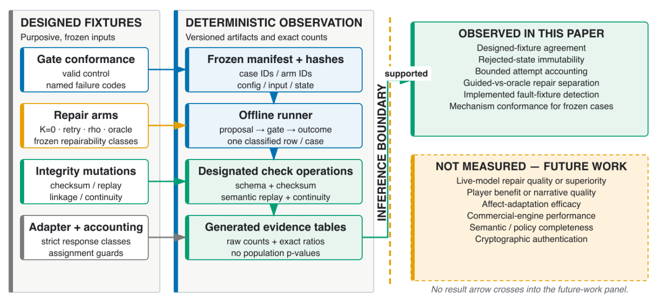


Fig. 5. Evidence boundary from frozen fixture through a designated check operation and generated table. Counts are admitted only through generated artifacts.

Fig. 5 makes the offline reporting boundary explicit: only rows admitted by the frozen fixture and designated check operation enter the offline generated tables. The live table follows a separate path through the hash-bound screening promotion manifest; its rows never enter the frozen offline denominator.

ENG1. Godot 4.7.1 replayed the retained *Sealed Lighthouse* canonical trace in a non-headless 1280×720 SubViewport. Its manifest binds each PNG to event/delivery indices, validation, and state hashes: Fig. 4(b) preserves the before hash, whereas (c) changes it after a valid commit. All source panels are generated-asset-free primary-track renders; the illustrative crops below enlarge only their authored status/event callouts.

The selected v5 packet establishes one scripted fixture's render/state correspondence only; it adds no live model or Python transport, participant or player input, adaptation intervention, or narrative-quality result.

VII. Discussion

A. Established Scope

The designed pilot can establish mechanism behavior only for its frozen cases: whether encoded checks isolate state, whether bounded repair and fallback follow their declared paths, whether named corruptions are rejected by their designated operations, and whether assigned rows remain classified for aggregation. It does not attribute a rejection to a stable typed detector code. This is useful for authoring-time validation, continuous-integration regression, recorded-adapter tests, and pre-engine replay inspection.

The guided-versus-oracle boundary is the honest reading of the offline repair comparison. The state-reading callback remains the upper bound: it may consult the authoritative state, so it converts the oracle-only case that ρ, by construction, cannot. The counterexample-guided operator matches the oracle exactly on the guided-repairable class while consuming only the validator's structured counterexamples. The live screening extension then probes whether that mechanism transfers to hosted proposals. It separated ρ from blind retry only when the constructed state caused every proposal to land in the guided-repairable class; already-valid and guided-irreparable regimes remained null. This regime dependence, together with the unpinned hosted revision and quasi-replicated calls, blocks a general sample-efficiency or model-capability claim. Feedback-consuming repair loops are precedented: Self-Refine feeds back language generated by the same untrusted model [44], and Self-Debugging consumes execution results from a trusted full-program runtime [45]. ρ differs in channel and authority: its feedback is a typed error set computed by the deterministic validator before execution, and every failed live arm remains state-isolated.

B. Relationship to Prior Work

TRACE-RPG complements constrained decoders by applying state-relative authorization after parsing [5], [6], [7]. It differs from retrieval and memory components by withholding commit authority [8], [9], [10]. It complements executable environments and benchmarks rather than replacing them [1], [2], [22], [25]. Process-linked traces instantiate a measurement concern emphasized by AgentBoard [23]; they are not evidence that TRACE-RPG outperforms that or any other system.

The studies that would test confirmatory efficacy—a preregistered multi-model comparison with an independently labelled semantic oracle and a genuine blind-retry comparator, retrieval and memory ablations at matched encoded validity, and a blinded matched-validity human study with justified power and mixed-effects analysis—remain outside this paper. The screening pilot does not substitute for any of them [22], [33], [30], [38].

VIII. Threats, Ethics, and Artifact Scope

The evidence has nine explicit boundaries. **L1:** encoded predicates cannot detect semantics omitted from structured fields. **L2:** one authored world and one constructed live-state variant do not establish cross-world, cross-policy, or scale generalization. **L3:** the state-reading callback is an oracle; offline classes and regime-dependent screening do not establish general repair quality or sample efficiency. **L4:** the single hosted proposer has an unpinned revision, no token accounting, and seed labels that do not control sampling; no live Python–Godot authorization round trip exists. **L5:** no participant or affect data exist. **L6:** unkeyed hashes provide integrity, not writer authentication or adversarial tamper resistance. **L7:** record consistency assumes one writer; concurrent-writer safety is untested. **L8:** engine evidence is one Godot fixture with startup-dominated samples that miss 16.7 ms, not a portability or real-time result. **L9:** selected fixtures and screening cells support no population, safety-rate, or model-ranking inference. Internal validity also depends on independent policy expectations and no fixture leakage; reproducibility depends on fixed software and artifacts.

The pilot processes no participants or personal telemetry. Future human and affect work requires applicable review, informed consent, compensation disclosure, minimization, retention/deletion rules, and opt-out. LLM judges, if used, remain auxiliary because both automated and crowd judgements require calibration and bias controls [33], [34], [35], [36].

IX. Conclusion

TRACE-RPG isolates authored candidate-event payloads from canonical mutation through an encoded symbolic commit gate and makes terminal outcomes auditable through linked records and state-semantic replay. The deterministic offline pilot establishes mechanism conformance on frozen authored fixtures. The separate live screening pilot observed a 5/5 versus 0/5 guided separation only in the constructed repairable regime and null results elsewhere, so it remains pilot-only. Cross-model superiority, player-experience benefit, retrieval or memory effectiveness, affect efficacy, and commercial-engine performance remain explicit confirmatory extensions.

Disclosure of AI Usage

Large language models assisted prose, code and test drafting, command orchestration, evidence review, result interpretation, simulated peer review, and citation-identity checks; a hosted model also produced the separately labelled live-screening candidates. Every reported count comes from deterministic runners or hash-bound receipts, with offline, live-screening, and engine evidence kept separate. The authors cross-checked claims against source artifacts and take full responsibility for the content.

Data and Code Availability

The repository contains the implementation, build sources, separated offline and live-screening packets, and `contribution-evidence-matrix.csv`, `reference-topic-crosswalk.csv`, and `experiment-evidence-matrix.csv`. The offline manifest binds 38 artifacts, 22 inputs, and 121 provenance rows (85 executed and 36 aggregate); the live promotion manifest binds 14 JSON/JSONL receipts and retains the superseded v1 diagnostic. The portable offline invocation contains no absolute clone path, and clean-Git input binding is re-established at tag `trace-rpg-guided-repair-inputs-20260821-v1`. No anonymized reviewer archive or DOI deposit is claimed.

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